INCH-POUND

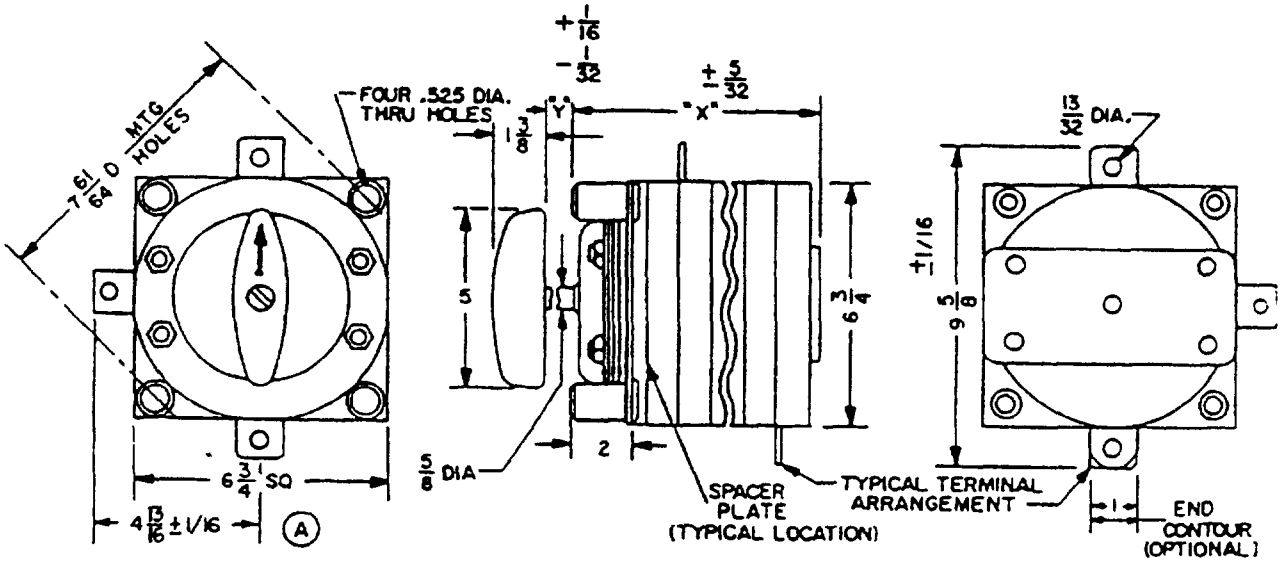
MIL-S-15291/12B(SH)
22 July 1991
SUPERCEDING
MIL-S-15291/12A(SH)
16 June 1986

MILITARY SPECIFICATION SHEET

SWITCHES, ROTARY, SNAP ACTION CLASS 20SR FRONT MOUNTED, SIDE CONNECTED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-15291.



NOTES
1. Dimensions are in inches
2. Unless otherwise specified, tolerance is plus or minus 0.015

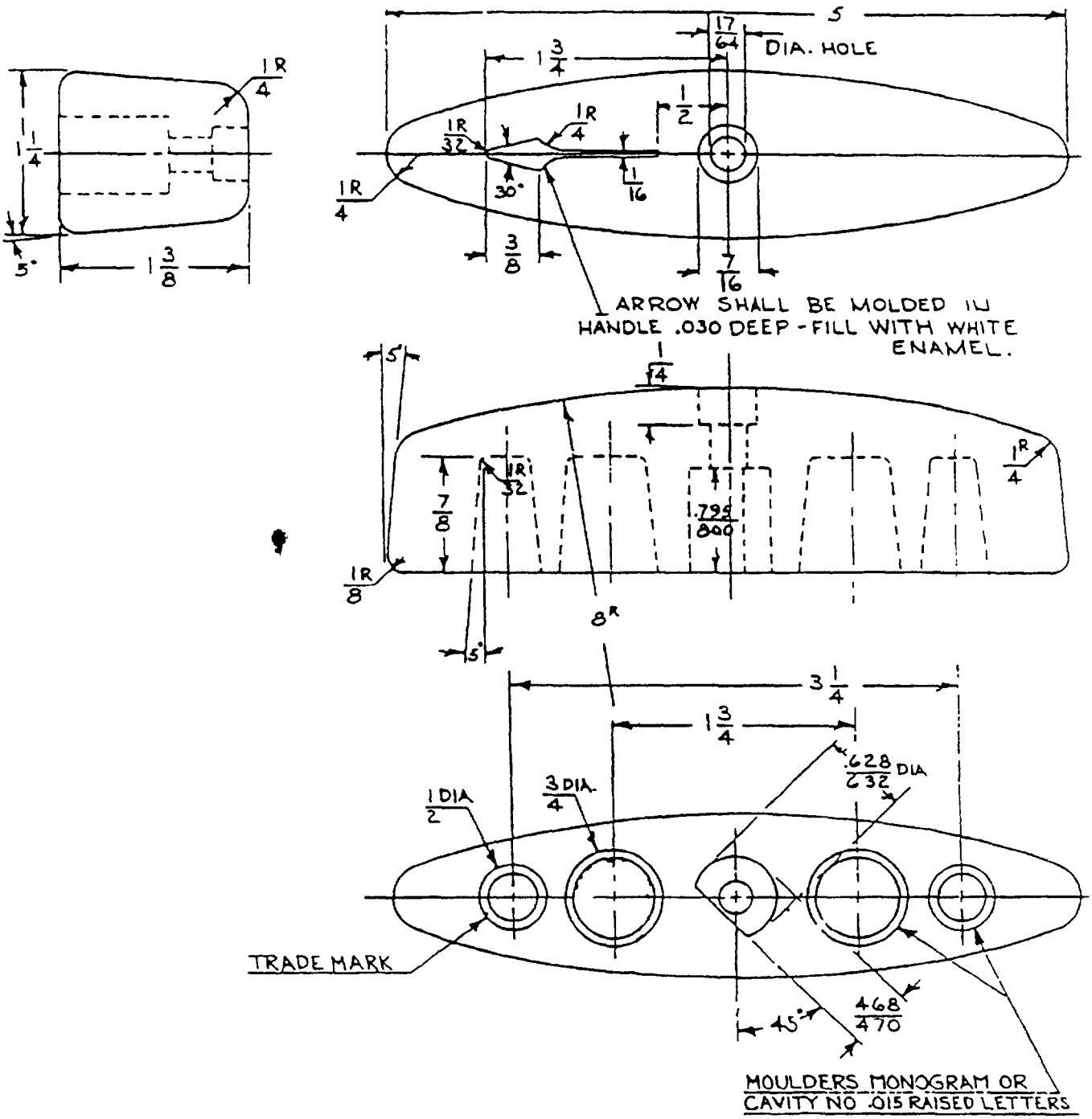
FIGURE 1 *Class 20SR switch*

AMSC N/A

FSC 5930

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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PIN M15291/12-100

FIGURE 2 *Handle.*

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TABLE I. *Type and switching characteristics*

MIL-S-15291/12 Dash No Type desig	Detail rel Torque in-lbs	Dim "X"	Dim "Y"	Handle or shaft	Circuit and spacer configuration rotor position, spacer locations, terminal marking and locations No 1	No 2	No 3	No 4	No 5	No 6	No 7	No 8	No 9	Notes
-001	Fig 1	4 13/16	7/16	OFF ON										
20SR2A2	72													
-002	Fig 1	5 11/16	7/16	OFF ON										
20SR3A2	72													
-003	Fig 1	6 9/16	7/16	OFF ON 1 2										
20SR2B2	72													
-004	Fig 1	8 5/16	7/16	OFF ON 1 2										
20SR3B2	72													
-005	Fig 1	6 9/16	7/16	OFF ON 1 3										
20SR2E2	72			ON 2										
-006	Fig 1	8 5/16	7/16	OFF ON 1 (ON)										
20SR3E2	72													
-007	Fig 1	7 7/16	7/16	1&2 OFF										
20SR4F2	72													
-008	Fig 1	9 3/16	7/16	1&2 OFF										
20SR6F2	80													
-009	Fig 1	4 13/16	5/16	ON 1 OFF 2										
20SR2S2	72													
-010	Fig 1	5 11/16	5/16	ON 1 OFF 2										
20SR3S2	72													

TABLE I *Type and switching characteristics*

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REQUIREMENTS:

Applicable specification:	MIL-S-15291
Dimensions and mounting:	See figure 1
Switching characteristics:	See table I
Angle of throw:	90 degrees
Switching action:	Snap action, reciprocating
Electrical and endurance ratings:	See table II
Stop strength (applicable to switches with stops):	100 inch-pounds
Vibration:	50 Hz, MIL-STD-167-1
Shock (high impact):	High impact, MIL-S-901
Insulation (spacers and decks):	Glass alkyd, MAI-60 (or MAI-30) in accordance with MIL-M-14
Spacer plate:	MAI-60, MAI-30, or CFI-10 in accordance with MIL-M-14
Contact resistance:	1.5 milliohms maximum
Dielectric withstanding voltage:	2,000 Vrms
Insulation resistance:	200 megohms minimum
Contact material:	Rotor, phosphor bronze – Stationary contact, copper
Mounting bolts	(4) 1/2 diameter, length, and headstyle to suit application. Bolts not furnished.

TABLE II. *Electrical and endurance ratings.*

Tests	Current (amperes)	Voltage (volts)	Electrical operations (number of operations)	Test rate (operations per minute)	De-energized operations (number of operations)
Alternating current (rms)	200	500	6,000	8	4,000
Direct current	200	250	4,000	3	-

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Terminal marking: Terminal markings shown in table I locate terminals as viewed from the front of the switch (handle end) Markings shall be stamped in rear surfaces of terminals.

GENERAL INFORMATION:

Switches not covered by specification sheets: Switches which are fabricated from standard parts, as used in qualified switches, but which do not comply with switches detailed herein with respect to circuit characteristics, switching action, mounting arrangement, and handle details may be acquired under this specification from contractors having qualification approval under this specification.

Extended ratings: Switches detailed herein have been tested and found satisfactory at the extended ratings listed in table III. Reduced life expectancy must be anticipated for switches used at these increased voltage or current levels. Tests under the conditions of table III are not required for qualification acceptance and they are not repeated routinely as for maintenance of qualification. Supplemental evaluations and tests applicable to particular circuit requirements are recommended.

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TABLE III. *Extended ratings.*

Switching characteristics	AC 60 or 400 Hz											
	125 volts				250 volts				500 volts			
	Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.		Resistive or lamp load		Inductive load 0.75 p.f.	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000	200	6,000
	DC											
	120 volts				250 volts				350 volts			
	Resistive or lamp load		Inductive load		Resistive or lamp load		Inductive load[1]		Resistive or lamp load		Inductive load	
	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations	Amp.	Operations
All	200	4,000	---	---	200	4,000	200	4,000	---	---	---	---

[1] 0.025 henry for inductive circuit

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APPLICATION AND ACQUISITION GUIDE: PIN and type designation cross reference shall be as shown in table IV

TABLE IV. *Application and acquisition guide.*

M15291/12 dash No.	Type designation	For new or existing design	For re-placement	Circuit configuration
-001	20SR2A2	X		A (off-on-off-on)
-002	20SR3A2	X		A (off-on-off-on)
-003	20SR2B2	X		B (off-on-1-off-on 2)
-004	20SR3B2	X		B (off-on-1-off-on 2)
-005	20SR2E2	X		E (off-on-1-on 2-on 3)
-006	20SR3E2	X		E (off-on-1-on 2-on 3)
-007	20SR4F2	X		F (off-on 1-on 1 & 2-on 2)
-008	20SR6F2	X		F (off-on 1-on 1 & 2-on 2)
-009	20SR2S2	X		S (on 1-on 2-off-off)
-010	20SR3S2	X		S (on 1-on 2-off-off)

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5930-N663-13)

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